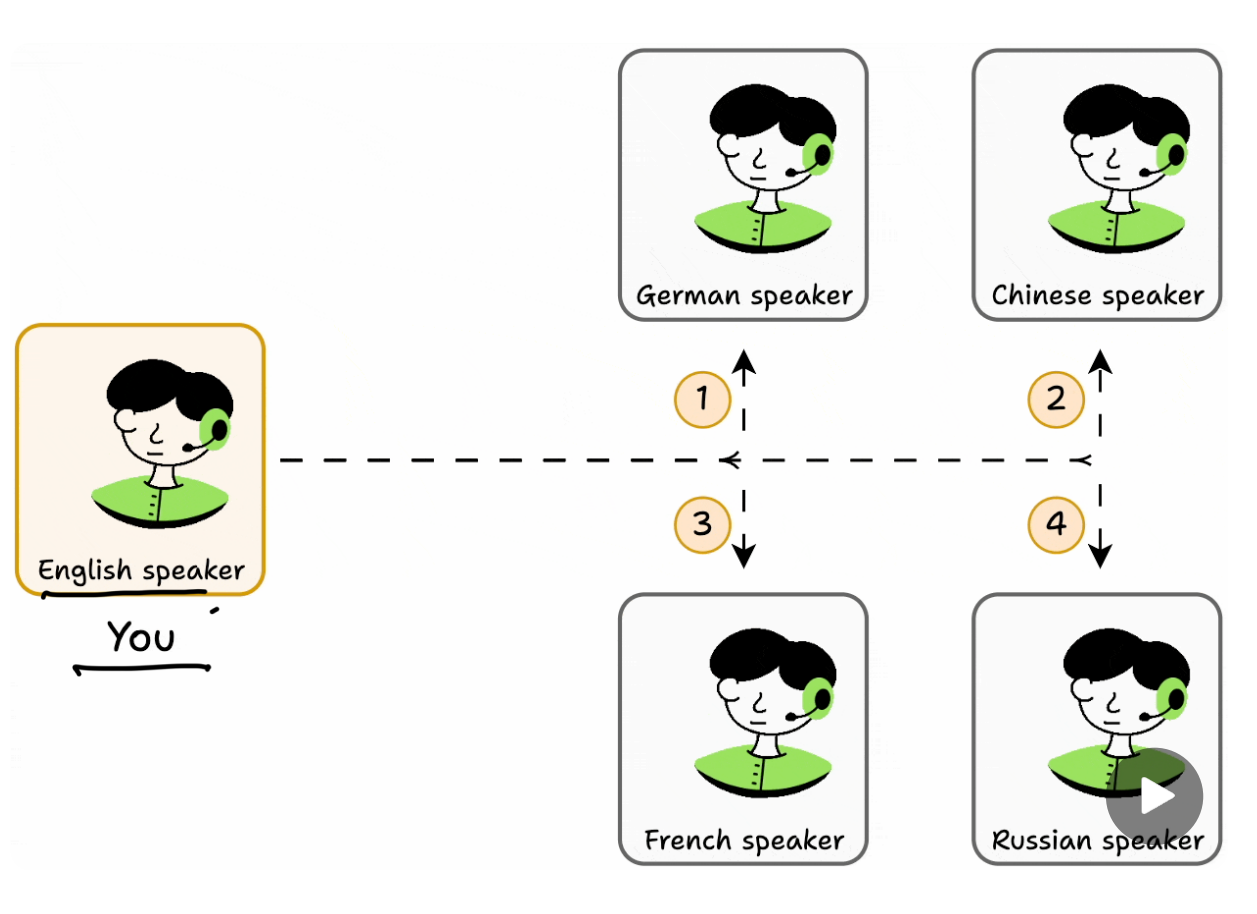
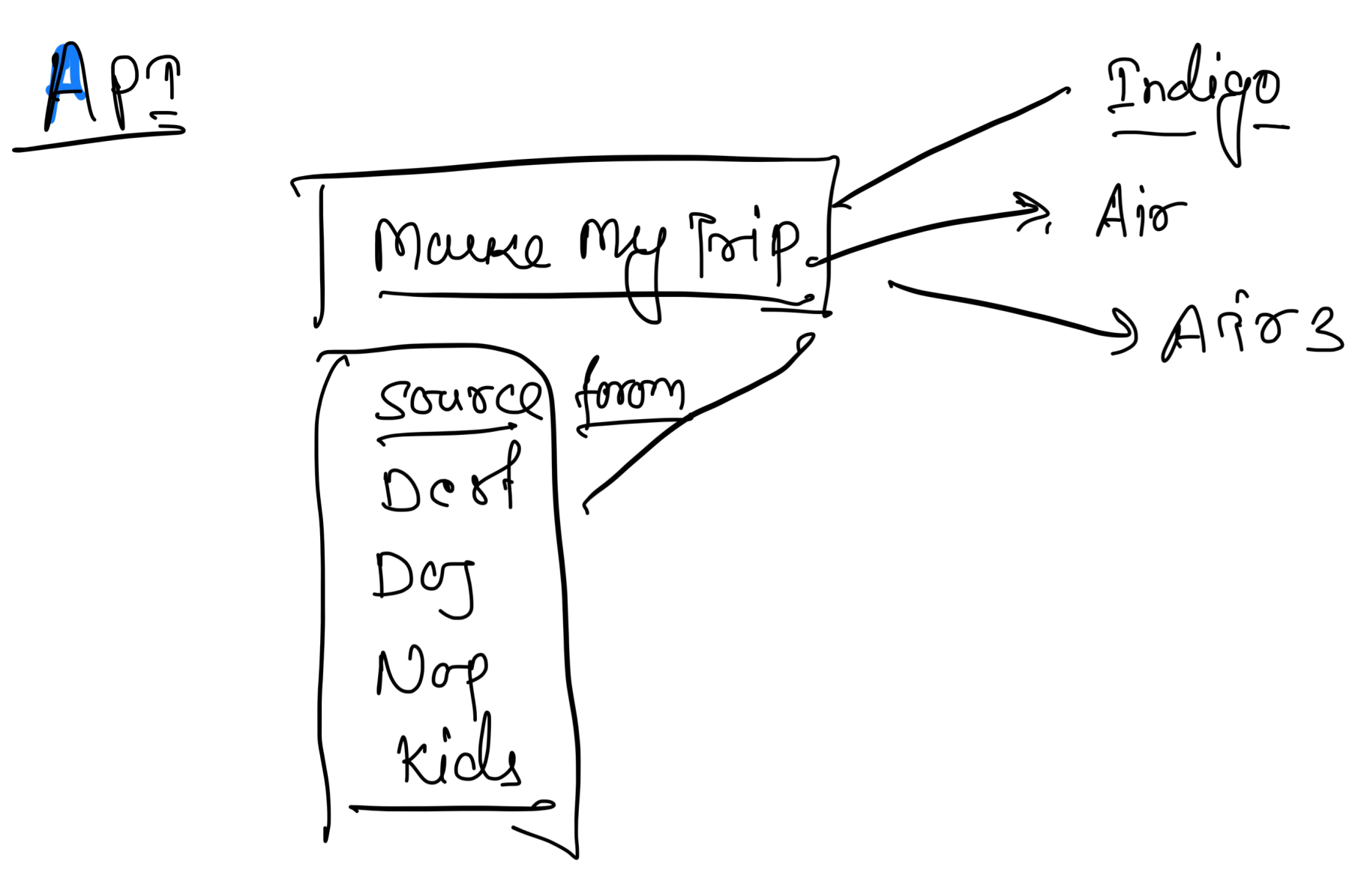
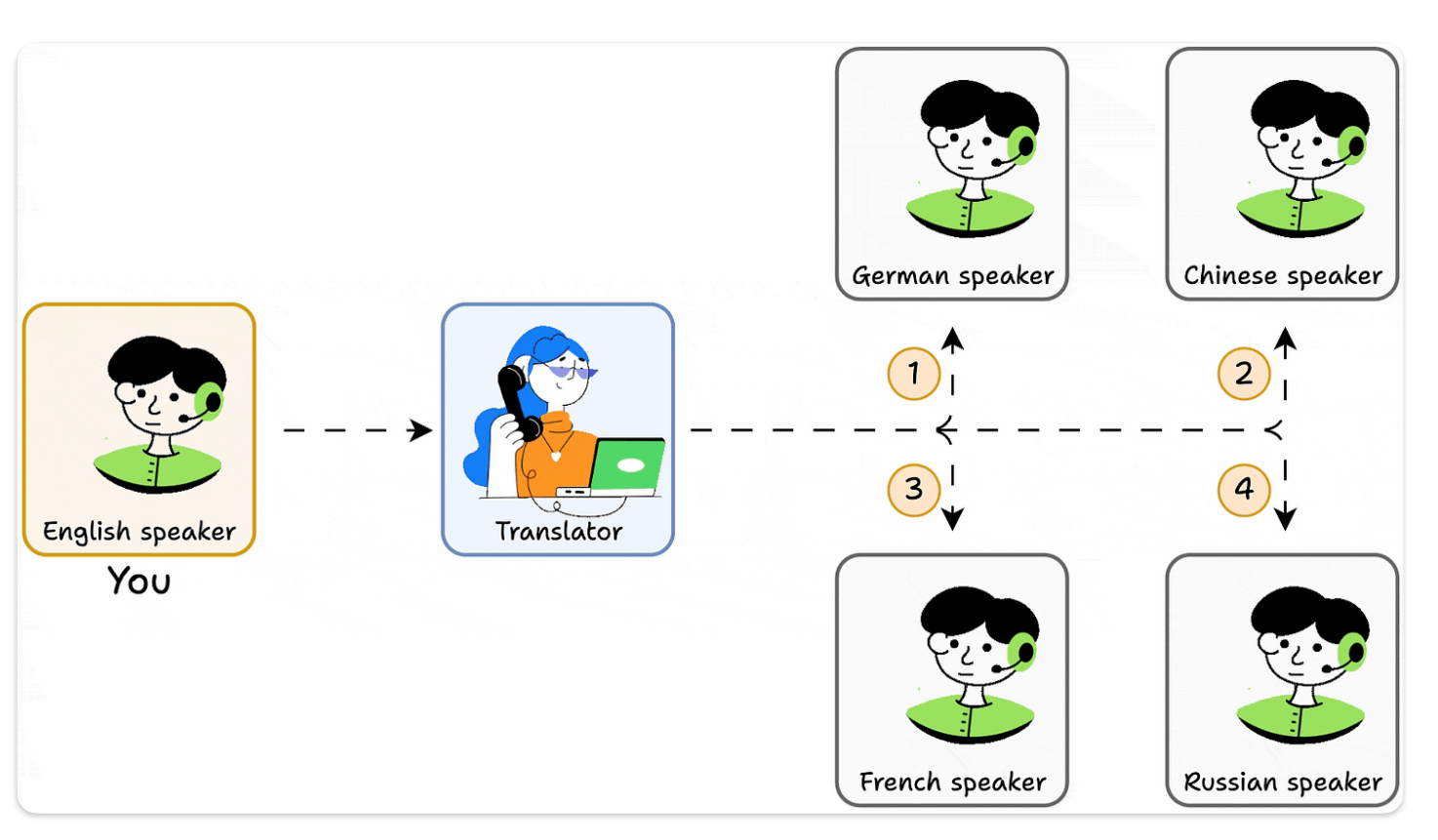
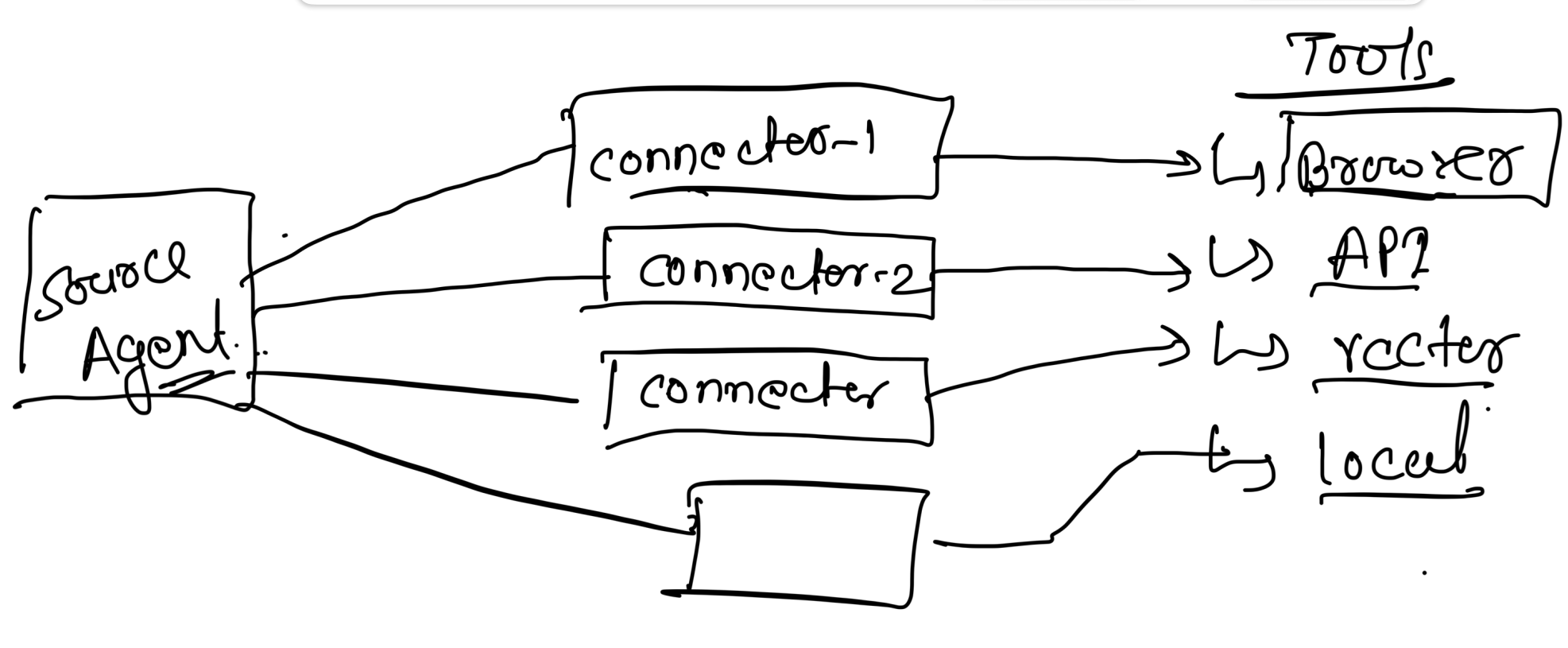
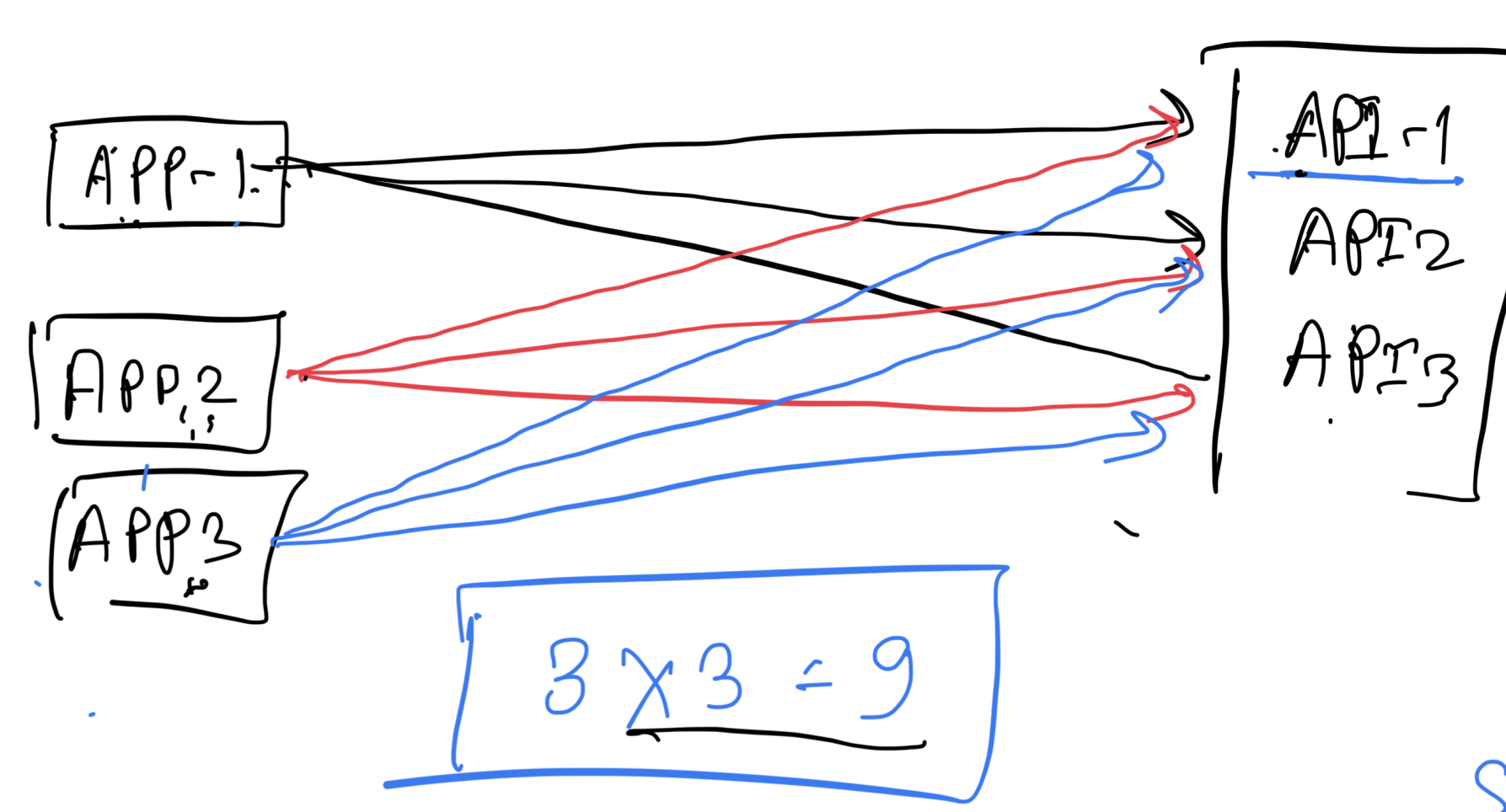
Model context protocol (mcp)

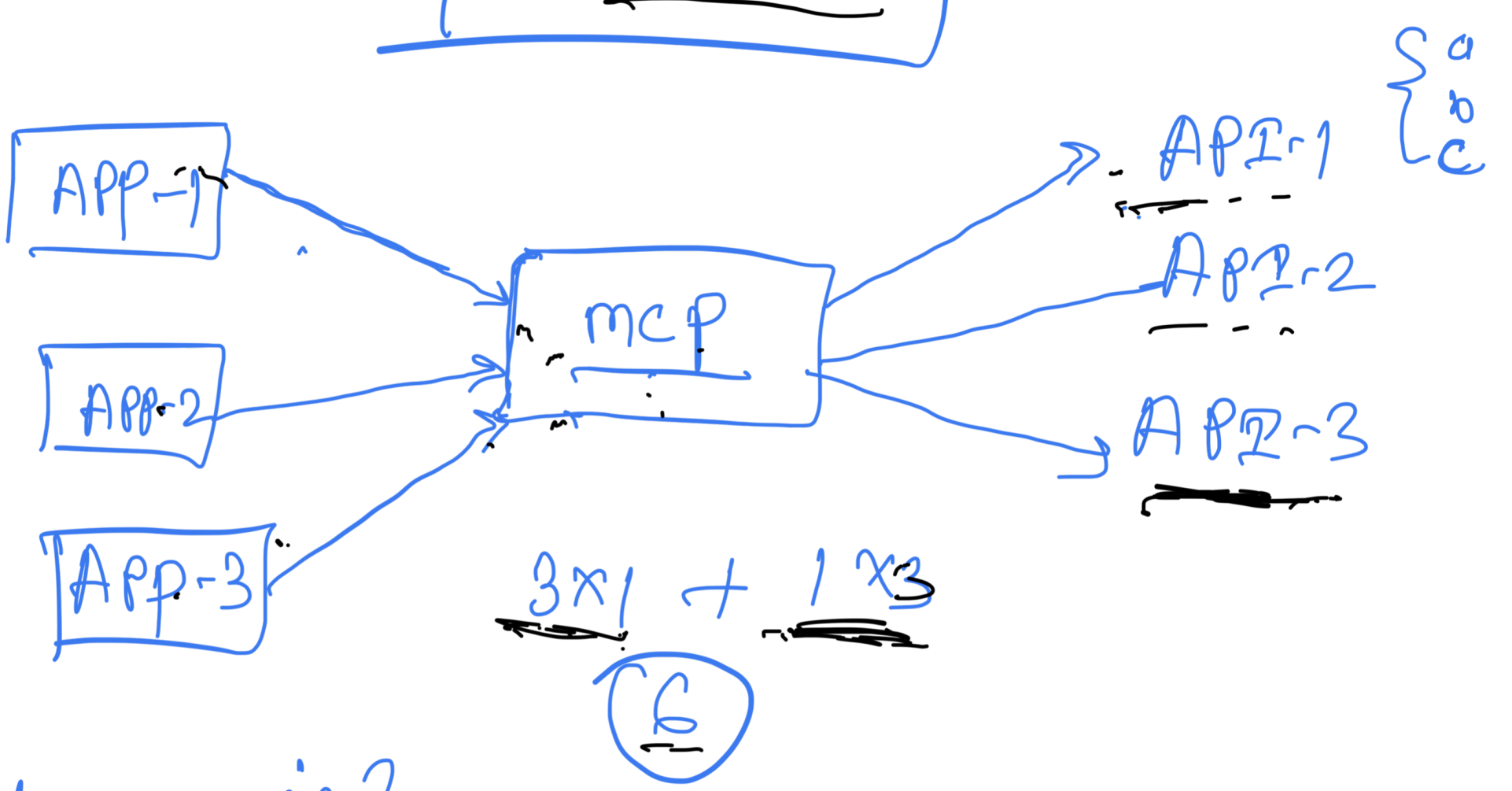

German speaker
Chinese speaker
English speaker
You
1
2
3
4
French speaker
Russian speaker


API
Make my Trip
Indigo
Air
Air3
Source
Dest
Day
Nop
Kids
from


English speaker
You
Translator
German speaker
Chinese speaker
1
2
3
4
French speaker
Russian speaker

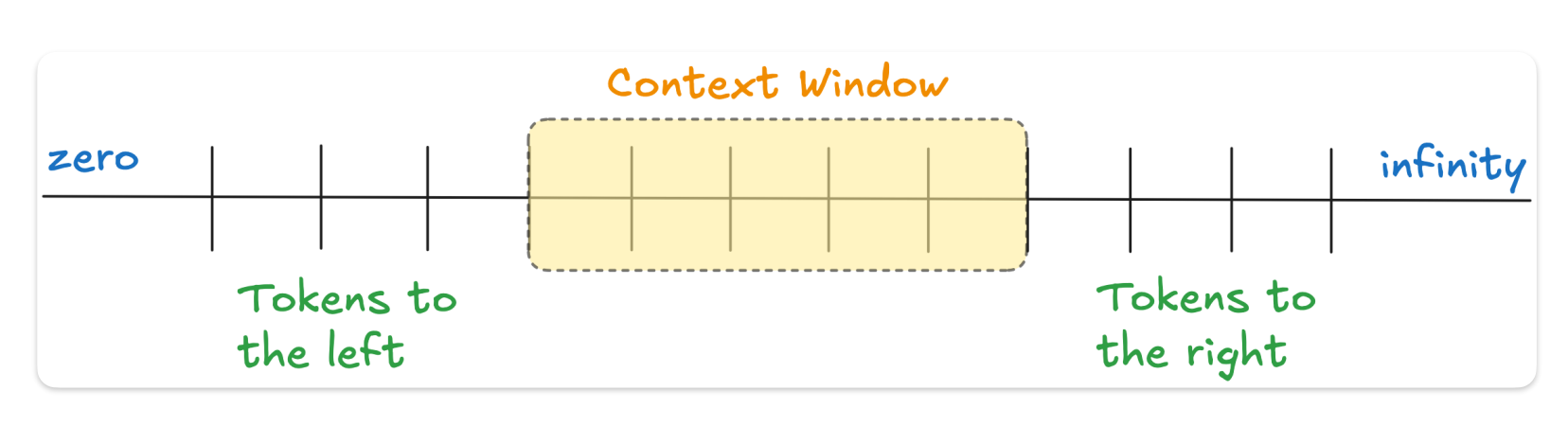

Source Agent
connector-1
connector-2
connector
Tools
Browser
API
vector
local

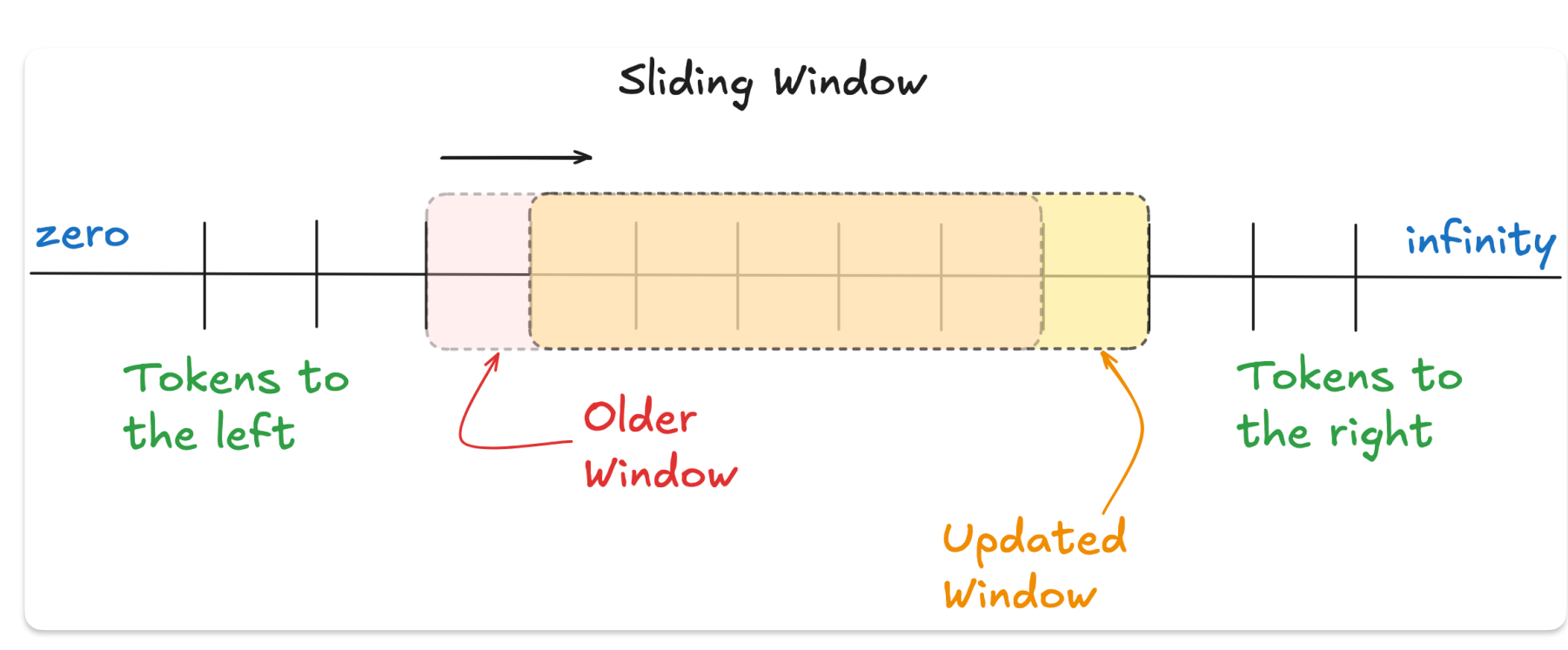

APP-1
APP2
APP3
API-1
API2
API3
3×3 = 9

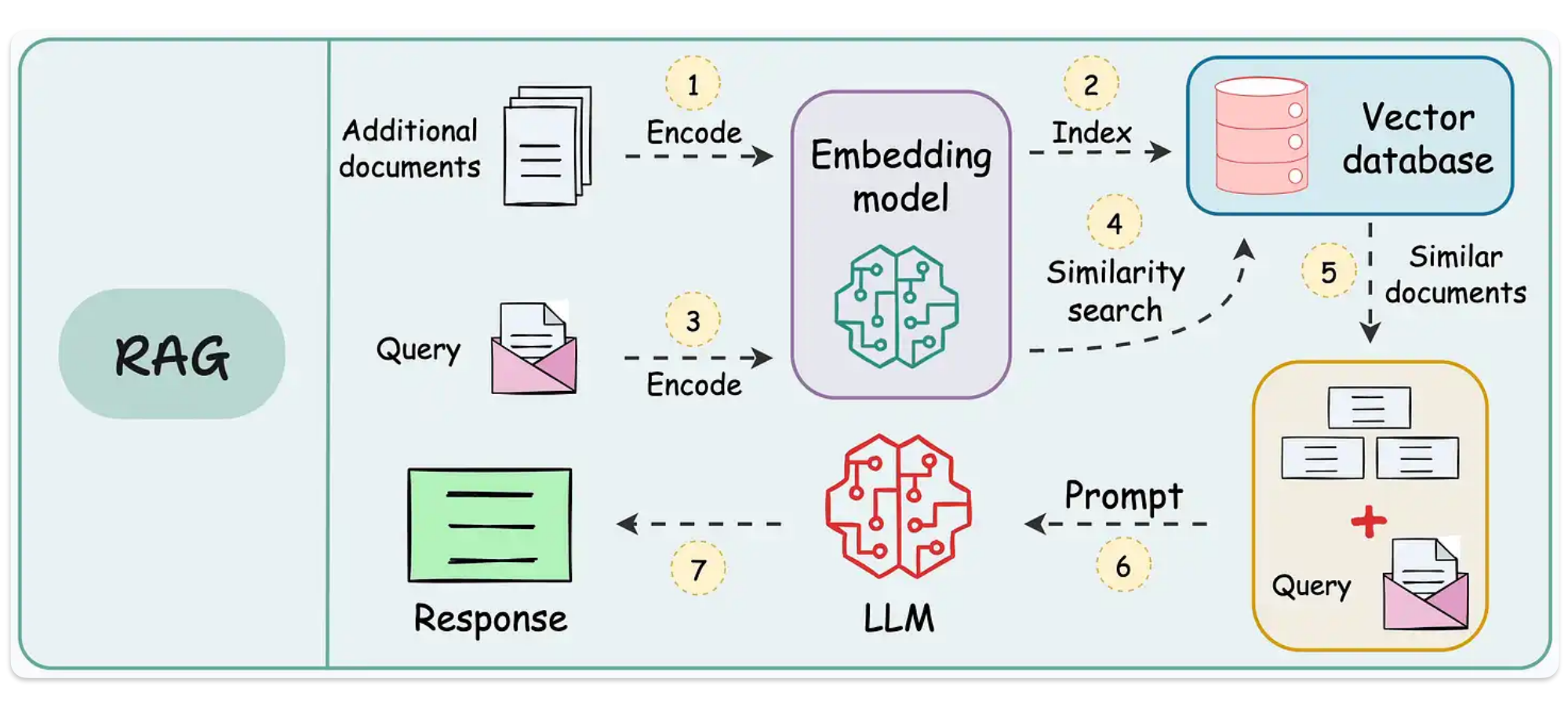

APP-1
APP-2
APP-3
mcp
API-1
API-2
API-3
S
a
b
c
3×1 + 1×3
6

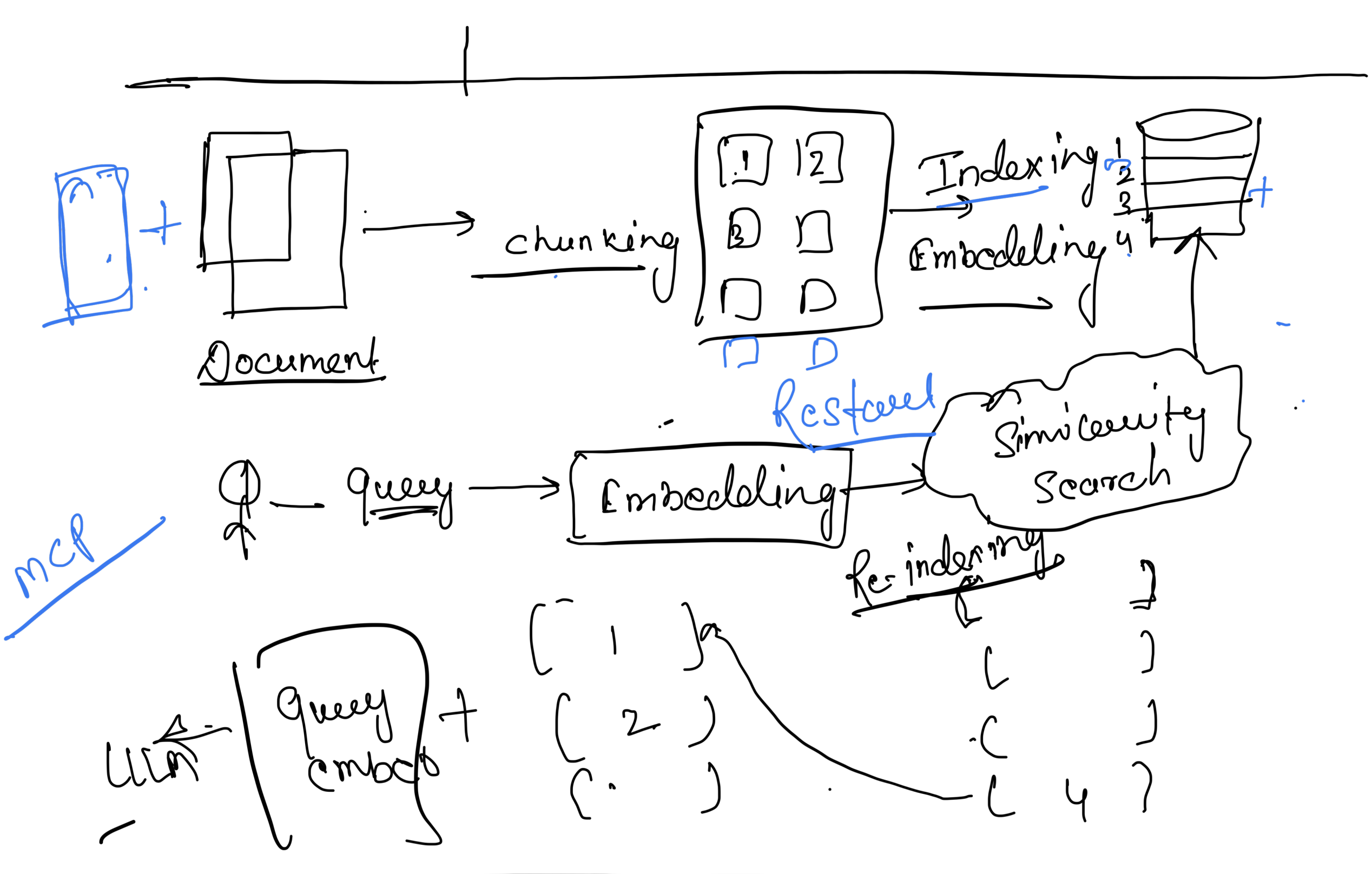
① what mcp is?
② How mcp enhances context-Awareness AI system
③ How to use mcp to dynamically provide AI system with necessary context.

Early approaches to context management


Context Window
zero
infinity
Tokens to the left
Tokens to the right

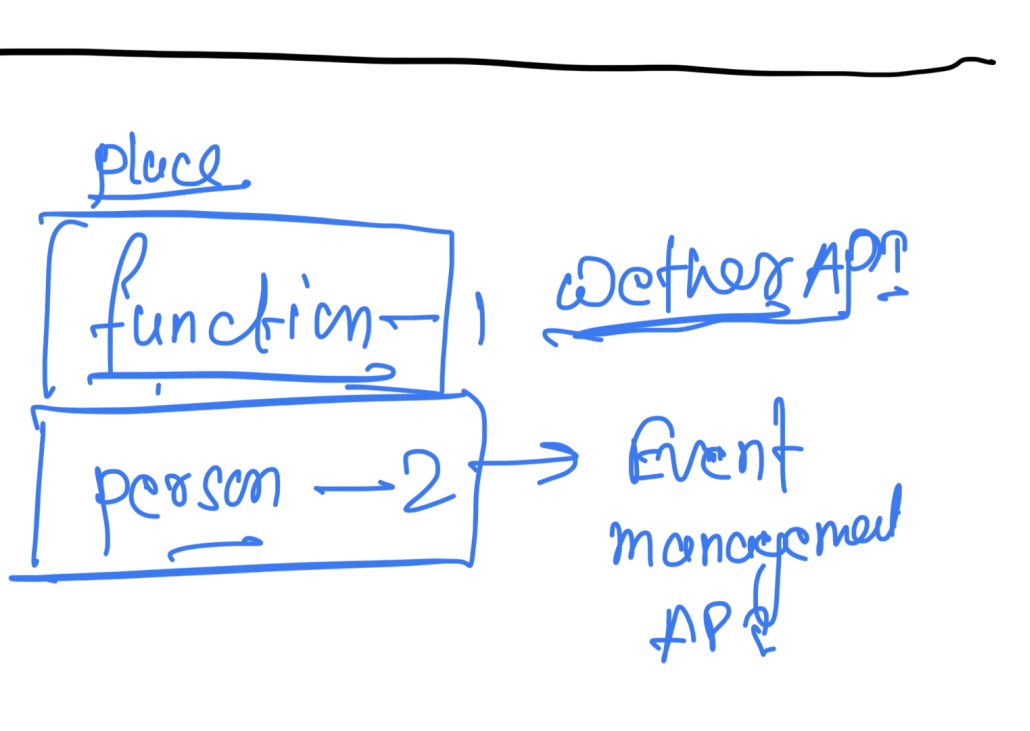

Sliding Window
zero
infinity
Tokens to the left
Older Window
Updated Window
Tokens to the right

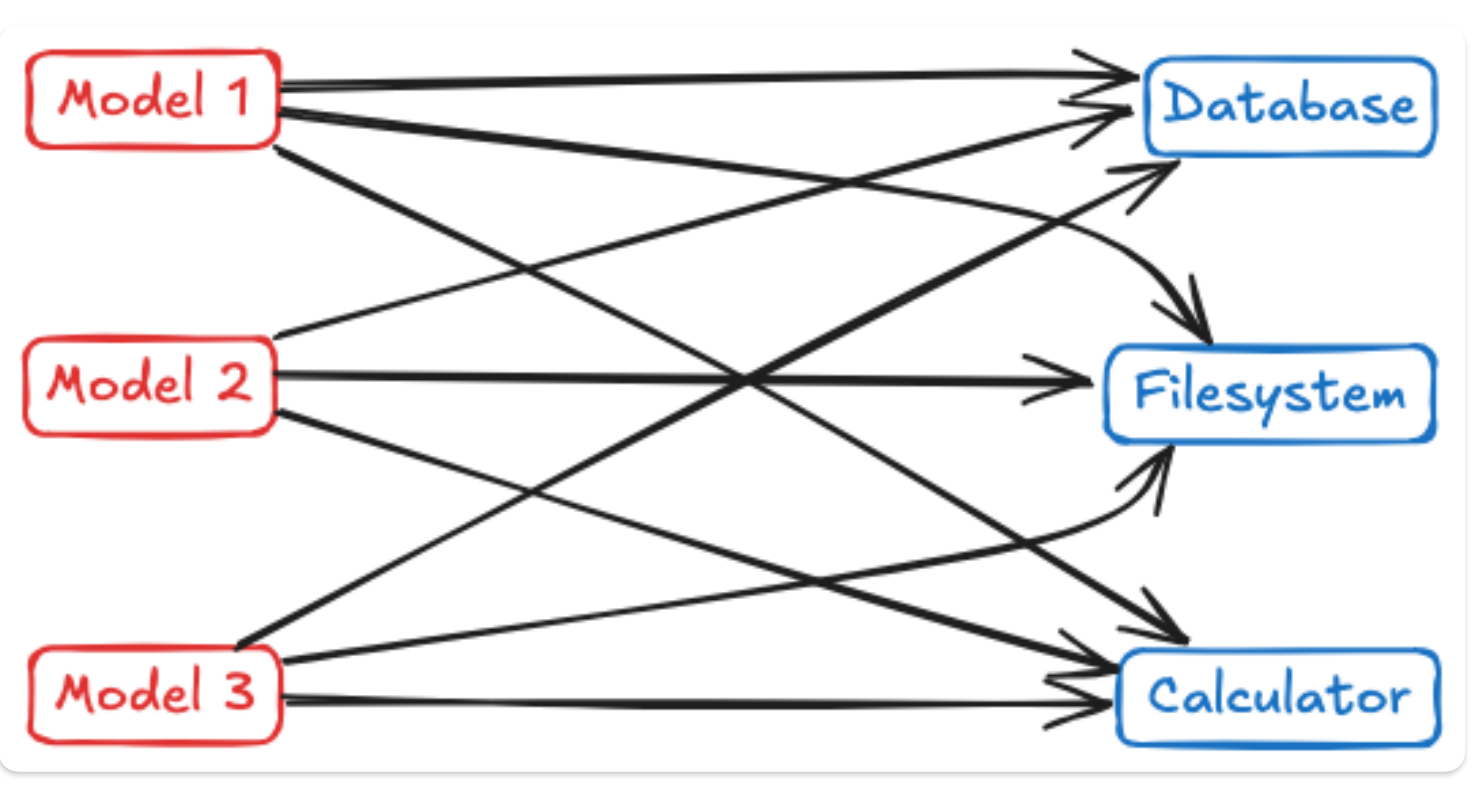

RAG
Additional documents
1
Encode
Embedding model
2
Index
Vector database
4
Similarity search
5
Similar documents
Query
3
Encode
Response
7
LLM
Prompt
6
Query


+
Document
chunking
1
2
3
Indexing
1
2
3
4
Embedding
Restored
Similarity Search
mcp
query
Embedding
Re-indexing
LLM
Query embed
+
1
2
4

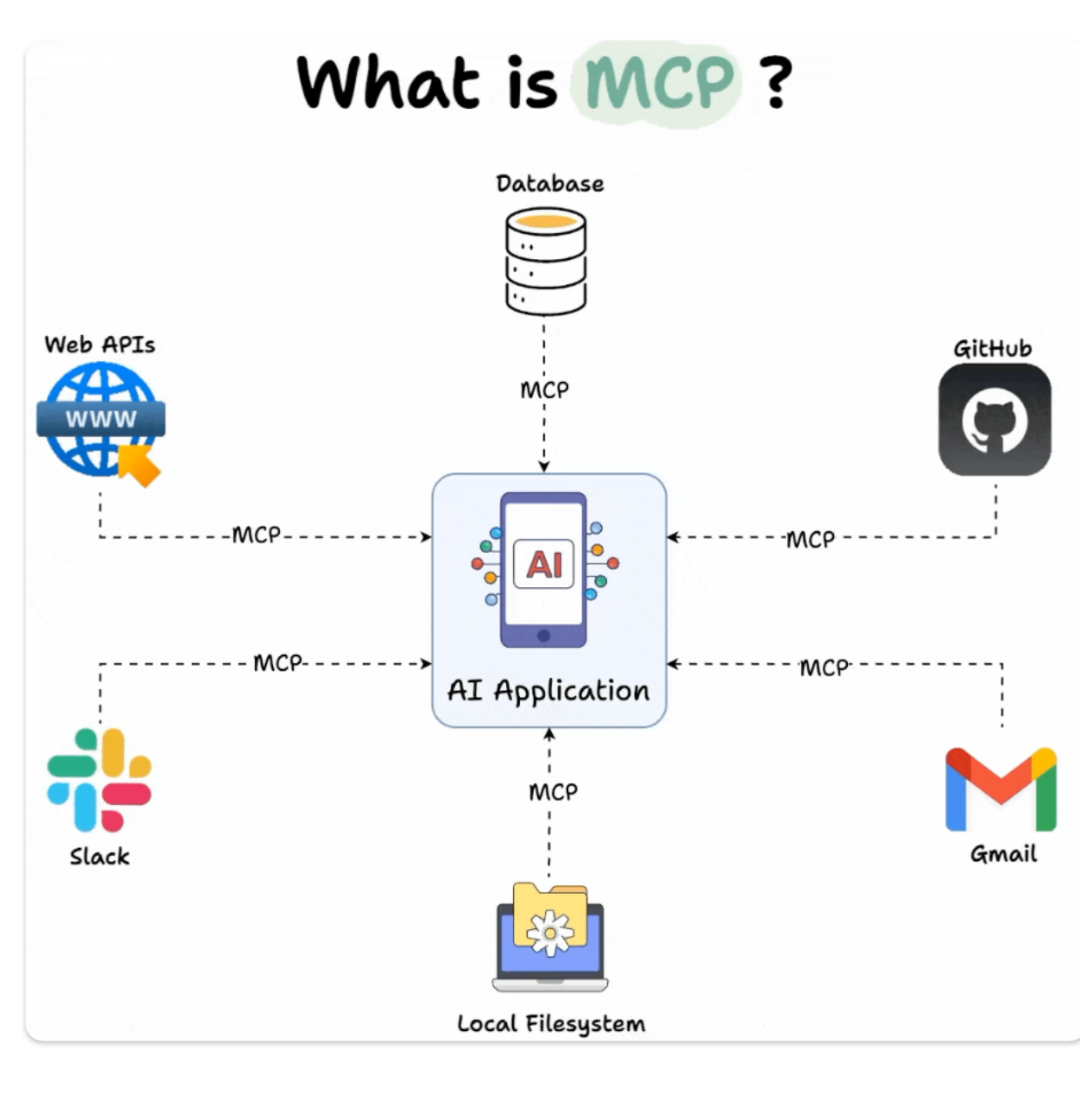
function calling / Tool calling

comple


place
function—1
Wether API
person —2
Event management API

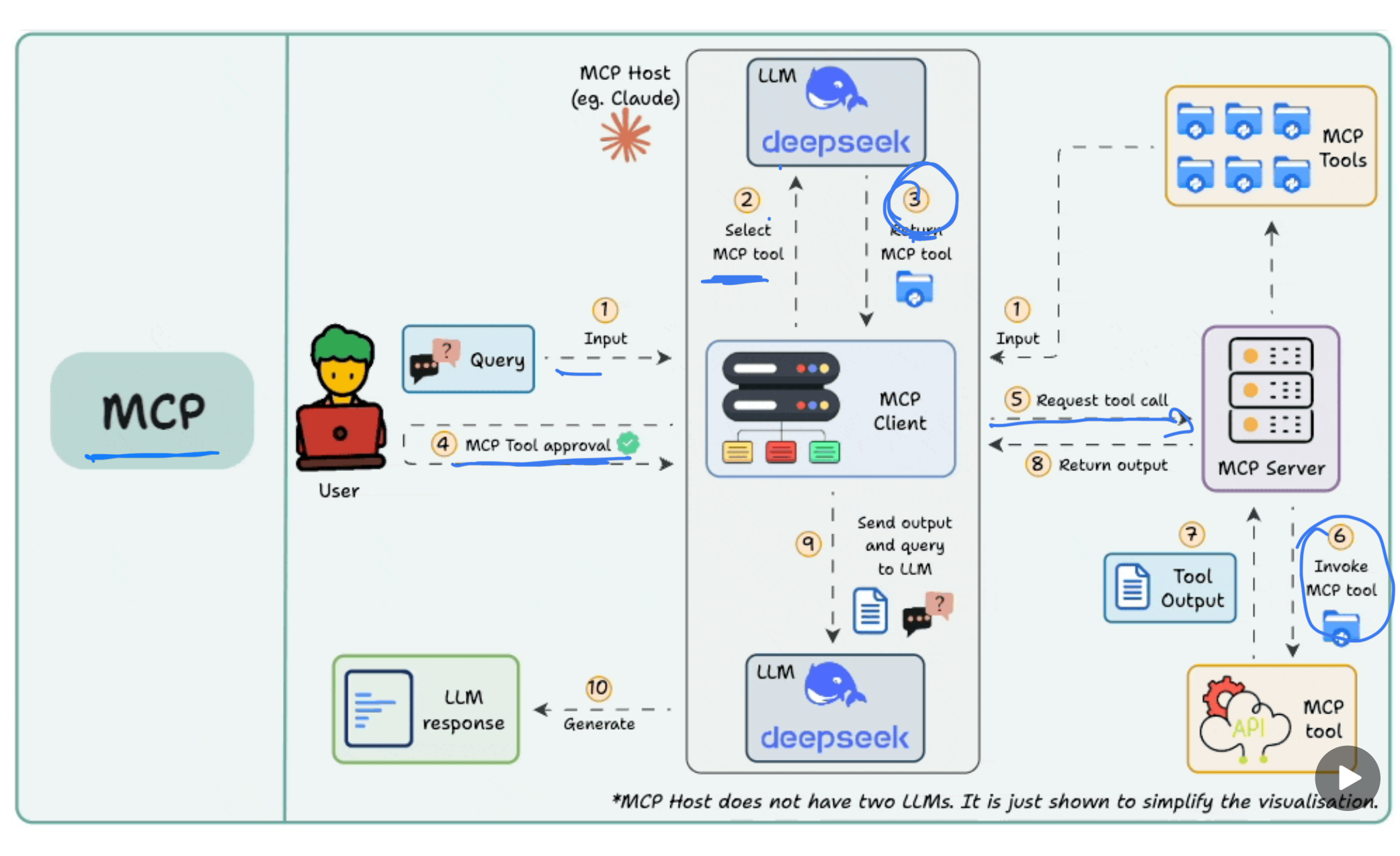

Model 1
Model 2
Model 3
Database
Filesystem
Calculator

m×n


What is MCP ?
Database
MCP
Web APIs
GitHub
MCP
MCP
AI Application
MCP
MCP
Slack
Gmail
MCP
Local Filesystem

Model context protocol (MCP) is a standardized interface and framework that allows AI models to seamlessly interact with external tools, resources, and environments.

Rayesh
↳ I am lead Data Sci.
↳ Data →
↳ BIA - 3 year


MCP
MCP Host (eg. Claude)
User
Query
1
Input
4
MCP Tool approval
LLM
deepseek
2
Select MCP tool
3
Return MCP tool
MCP Client
9
Send output and query to LLM
LLM
deepseek
10
Generate
LLM response
MCP Tools
1
Input
5
Request tool call
8
Return output
MCP Server
7
Tool Output
6
Invoke MCP tool
MCP tool
API

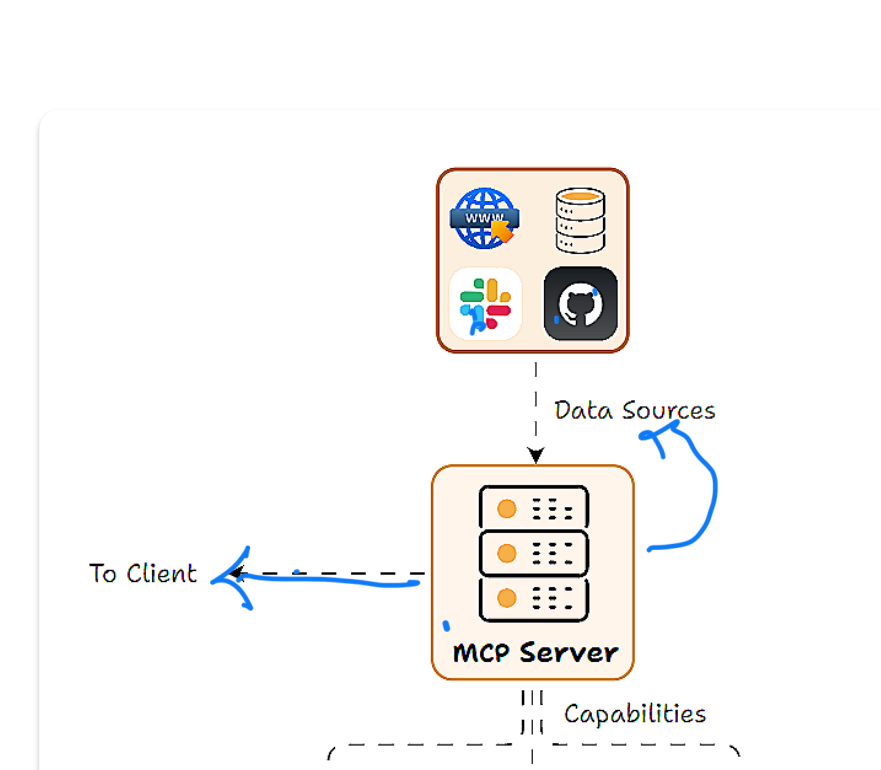
*MCP Host does not have two LLMs. It is just shown to simplify the visualisation.

①

Server

The MCP Server is the external program or service that actually provides the capabilities (tools, data, etc.) to the application.

AI App


Data Sources
To Client
MCP Server
Capabilities
Tools
Resources
Prompts

mcp servers consist of 3 major components

① Tools :— @mcp.tool.
add
sub
mul
div

②

```
Tools are what they sound like: functions that do something on behalf
of the AI model. These are typically operations that can have effects
or require computation beyond the AI’s own capabilities.
```

```
@mcp.tool()
def get_weather(location: str) -> dict:
    """Get the current weather for a specified location."""
    # (In a real implementation, call an external weather service)
    return {
        "temperature": 72,
        "conditions": "Sunny",
        "humidity": 45
    }
```

② # **Resources**

Resources provided read only data to AI

~~3~~
tool1 → [Resources]
too2
too3

③ prompts

prompts :— prompt in mcp to make context aware about the tools of mcp

Server

```
@mcp.prompt()
def code_review(language: str) -> list:
    """Provide a structured prompt for reviewing code in the given
language."""
    return [
        {"role": "system", "content": f"You are a meticulous
{language} code reviewer..."},
        {"role": "user", "content": f"Please review the following
{language} code:"}
    ]
```

Mcp Transportation Mechanism

① Stdio (client/server) → protocol

```
This transport is used when the Server runs as a local subprocess on
the same machine as the Client/Host.
```

300

API → mcp serve

Backer

Form